Devixo Solutions

ADVANCED EMPLOYEE MANAGEMENT SYSTEM WITH PAYROLL

C++ Programming Internship

TASK 04 — PROJECT REPORT

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Internship Domain: C++ Programming

Organization: Devixo Solutions

Project Type: Console-Based Application

Programming Language: C++

Development Environment: Code Utility Online C++ Compiler

Submission Date: 31 October 2026

Project Documentation and Implementation Report

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INTRODUCTION

The Advanced Employee Management System with Payroll is a console-based application developed in C++ as part of Devixo Solutions Internship Task 04.

This project demonstrates advanced C++ programming concepts through an interactive Employee Management System with Payroll. It manages full-time and part-time employee records, including adding, displaying, searching, updating, deleting, and sorting employees.

The system calculates gross and net salaries, generates payroll summaries and individual salary statements, and maintains payroll history through file handling.

It applies OOP concepts, inheritance, polymorphism, STL vectors, exception handling, input validation, and file I/O, providing practical experience in developing a structured C++ application.



Inclusivity



Transparency



Accountability

PROJECT OBJECTIVE

The primary objective of this project is to develop an Advanced Employee Management System with Payroll using C++.



The specific objectives are:

- To create a menu-driven console application for employee management.
- To implement employee addition, display, search, update, and deletion operations.
- To demonstrate classes, objects, encapsulation, and constructors.
- To implement inheritance and polymorphism through different employee categories.
- To manage employee records using STL vectors.
- To implement searching and sorting functionality.
- To calculate employee gross pay and net pay using salary-related information.
- To generate individual salary statements and maintain payroll history.
- To apply input validation and exception-handling techniques.
- To use file handling to save and retrieve employee records.
- To organize the application into modular functions and classes.
- To test the program's functionality and verify its behavior.

PROBLEM STATEMENT AND SCOPE

Problem Statement

Organizations need an organized way to maintain employee information, calculate salaries, manage payroll-related records, and retrieve employee details when required.

Managing these operations manually can become time-consuming and may lead to inaccurate records, difficulties locating employee information, and errors when calculating salaries or deductions.

The objective of this project is to address these basic requirements by developing a C++ application that manages employee records and performs payroll calculations through a console-based interface.

The system organizes employee information into individual records and provides operations for managing employees, searching and sorting records, calculating payroll, generating salary statements, and recording payroll history.

It also uses file handling to save employee information and payroll-related records, provided the relevant files can be accessed successfully.



Project Scope and Features

The application includes the following features:

1. **Add Employee**

Allows the user to create a new employee record by entering the required information, including employee ID, name, department, and the relevant salary details.

2. **Display Employees**

Displays the employee records maintained by the application so that users can review employee information.

3. **Search Employee**

Supports searching for employees using an employee ID or name, according to the available search options.

4. **Update Employee**

Allows the user to modify information associated with an existing employee record.

5. **Delete Employee**

Removes an employee record from the system when the requested employee exists.

6. **Full-Time Employee Management**

Supports employees whose pay is calculated using a monthly salary, bonus, and deductions.

7. **Part-Time Employee Management**

Supports employees whose base pay is calculated using an hourly rate and hours worked, with bonus and deductions applied according to the implemented formula.

8. **Payroll Calculation**

Calculates gross pay and net pay based on the employee's category and salary information.

9. **Sorting and Searching**

Provides operations for locating employees and organizing records using supported sorting criteria, including name and net salary.

Project Scope and Features

The application includes the following features:

10. **Payroll Summary**

Displays a summary of payroll information calculated by the application.

11. **Salary Statement Generation**

Generates an individual employee salary statement in a text file when the relevant operation succeeds.

12. **Payroll History**

Records payroll processing information in payroll_history.txt.

13. **File-Based Data Storage**

Uses employees.txt to save and load employee records and additional text files for payroll-related information.

14. **Menu-Driven Interface**

Provides an organized console menu through which users select and perform employee management and payroll operations.

TECHNOLOGIES USED

The project was developed using C++ and its standard library facilities.

1

Programming Language

C++
C++ was used to implement the application's logic, define employee classes, manage employee records, perform payroll calculations, organize searching and sorting operations, and handle file operations.

2

Development Environment

CodeUtility Online C++ Compiler
The application was developed and tested using an online C++ environment that provides a source-code editor and execution console without requiring a local compiler installation.

3

Header Files and Libraries

The program uses standard C++ headers appropriate to its implementation.

- iostream — for console input and output.
- string — for employee names, departments, and other text.
- vector — for maintaining a dynamically sized collection of employees.
- fstream — for reading and writing employee and payroll files.
- memory — for smart pointers used to manage employee objects.
- algorithm — for sorting employee records.
- Other standard headers required for input processing and calculations.

4

Programming Concepts

The implementation demonstrates:

- Classes and objects.
- Encapsulation.
- Inheritance.
- Polymorphism and virtual functions.
- Constructors and destructors.
- STL vectors and smart pointers.
- Searching and sorting.
- Functions and modular programming.
- Payroll calculations.
- File handling and data persistence.
- Input validation and exception handling.
- Menu-driven programming.

These concepts help organize the program and make its functionality easier to understand and maintain.

IMPLEMENTATION METHODOLOGY

The application was developed in stages so that its main components and operations could be implemented and tested systematically.

Step 1: Define the Employee Base Class

An abstract class named Employee provides the common employee interface and shared behavior. It defines the functions required for displaying employee information and calculating payroll.

Step 2: Implement the FullTimeEmployee Class

The FullTimeEmployee class inherits from Employee and represents employees paid using a monthly salary structure. It implements the payroll behavior associated with full-time employees.

Step 3: Implement the PartTimeEmployee Class

The PartTimeEmployee class inherits from Employee and represents employees whose base pay depends on an hourly rate and hours worked.

Step 4: Create the Employee Manager

The EmployeeManager class manages the employee collection and provides operations for adding, displaying, searching, updating, deleting, and sorting employee records.

Step 5: Implement Payroll Calculations

The program calculates gross pay and net pay using the employee's category and salary information.

Step 6: Implement Searching and Sorting

The application provides operations to locate employee records and sort them using the supported criteria.

IMPLEMENTATION METHODOLOGY

Step 7: Implement Payroll Statements and History

The program generates an individual salary statement and records payroll transactions in a history file when the relevant operations succeed.

Step 8: Implement File Handling

The application saves and loads employee information through employees.txt and maintains payroll-related records through separate text files.

Step 9: Implement Input Validation

The program checks user input according to its implemented validation rules, including numeric inputs and employee record requirements.

Step 10: Develop the Menu System

The main execution function displays the available operations and processes the user's selections until the exit option is selected.

Step 11: Test the Application

The program is tested through the interactive console. Employee records, payroll calculations, searching, sorting, file operations, and input validation are examined.

This staged methodology helps organize development and makes individual features easier to understand, test, and maintain.

PROGRAM STRUCTURE AND DATA REPRESENTATION

The application uses an object-oriented design in which each class has a specific responsibility.

1. **Employee Class**

The Employee class serves as the abstract base class for employee-related functionality. It defines the common interface for employee information and payroll behavior.
Virtual functions allow derived classes to provide their own implementations of payroll-related operations.

2. **FullTimeEmployee Class**

The FullTimeEmployee class inherits from Employee and represents employees paid through a monthly salary structure.
It stores the relevant salary information, including monthly salary, bonus, and deductions, and provides the payroll calculations associated with this employee category.

3. **PartTimeEmployee Class**

The PartTimeEmployee class inherits from Employee and represents employees paid according to an hourly rate and hours worked.
It implements the relevant base-pay calculation and applies bonus and deductions according to the program's payroll formula.

4. **EmployeeManager Class**

The EmployeeManager class coordinates employee management operations. Its responsibilities include maintaining employee records, searching for employees, updating and deleting records, sorting the collection, calculating payroll summaries, and handling file operations.

PROGRAM STRUCTURE AND DATA REPRESENTATION

5. **Vector-Based Storage**

The application uses:

vector<shared_ptr<Employee>>

This structure maintains a dynamically sized collection of employee objects through the common Employee interface.

Smart pointers help manage the lifetime of dynamically allocated employee objects and reduce the risk of manual memory-management errors.

6. **Function-Based Organization**

Separate member functions handle employee management, payroll processing, searching, sorting, data storage, and reporting.

This organization improves readability and makes the code easier to maintain.

7. **Main Function**

The main() function creates or initializes the employee manager, displays the main menu, and processes user selections.

Employee base class





Virtual Functions

Code Screenshots:











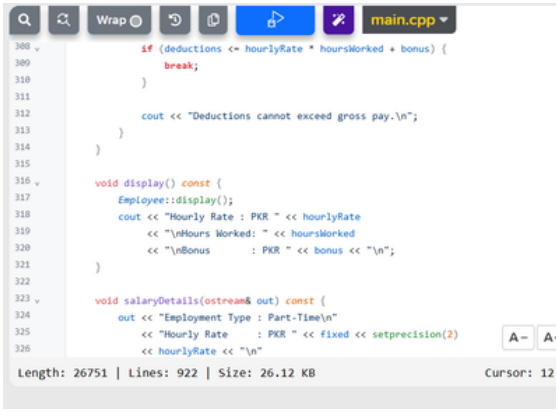


Input Validation Functions

Code Screenshots:











IMPLEMENTATION AND CODE EXPLANATION

The program is organized into separate functions to improve readability, testing, and maintainability.

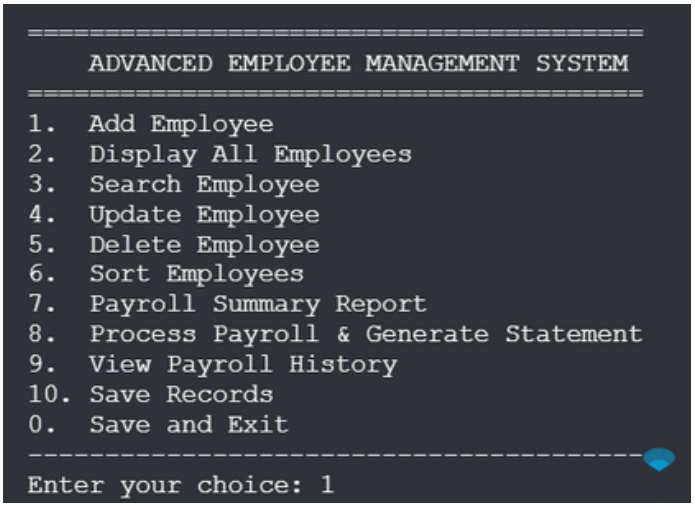
1. **Add Employee Function**

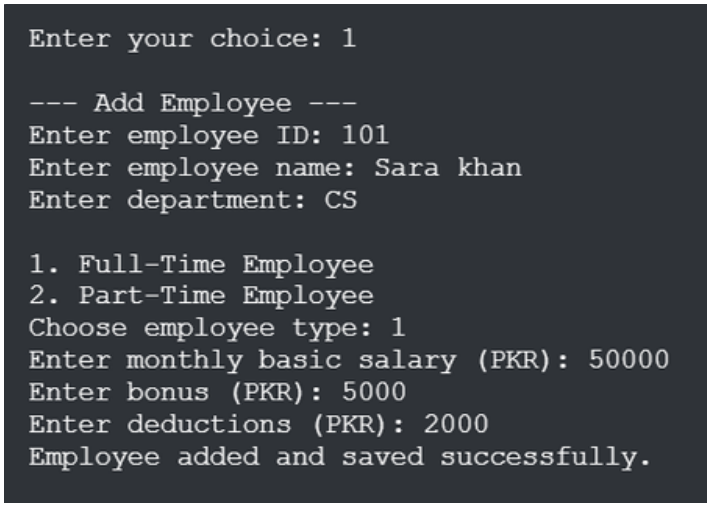
The employee-addition function collects the required information and creates a full-time or part-time employee record according to the selected category.

The program applies its employee ID validation and insertion logic before adding the record to the employee collection.

2. **Display Employees Function**

The employee-display function presents the employee records maintained by the application. It allows users to review employee IDs, names, departments, categories, and salary-related information.





IMPLEMENTATION AND CODE EXPLANATION

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 2
```

```
Employee ID : 102
Name        : Zeeshan Bashir
Department  : Mathematics
Type        : Part-Time
Gross Pay   : PKR 14000.00
Deductions  : PKR 100.00
Net Pay     : PKR 13900.00
Hourly Rate : PKR 1500.00
Hours Worked: 6.00
Bonus       : PKR 5000.00
----------------------------------
```

3. Search Employee Function

The search functionality locates employees using the supported search criteria, such as employee ID or name.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 3
```

```
--- Search Employee ---
1. Search by ID
2. Search by name
Choose search method: 2
Enter full or partial name: Zeeshan

Employee ID : 102
Name        : Zeeshan Bashir
Department  : Mathematics
Type        : Part-Time
Gross Pay   : PKR 14000.00
Deductions  : PKR 100.00
Net Pay     : PKR 13900.00
Hourly Rate : PKR 1500.00
Hours Worked: 6.00
Bonus       : PKR 5000.00
```

4/ Update Employee Function

The employee-update function locates an existing employee and modifies the supported fields according to the user's input.

IMPLEMENTATION AND CODE EXPLANATION

5. Delete Employee Function

The employee-deletion function removes the selected employee from the collection when the requested record exists.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 5
```

```
----------------------------
Enter your choice: 5

--- Delete Employee ---
Enter employee ID: 105
Employee not found.
```

6. Payroll Calculation Function

The payroll functionality calculates gross pay and net pay based on the employee category and the salary components stored in the employee record.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 8
```

```
--- Process Payroll ---
Enter employee ID: 102

Employee ID : 102
Name        : Zeeshan Bashir
Department  : Mathematics
Type        : Part-Time
Gross Pay   : PKR 14000.00
Deductions  : PKR 100.00
Net Pay     : PKR 13900.00
Hourly Rate : PKR 1500.00
Hours Worked: 6.00
Bonus       : PKR 5000.00
Type YES to record this payroll: yes
Salary statement saved as salary_statement_102.txt
Payroll recorded successfully.
```

IMPLEMENTATION AND CODE EXPLANATION

7. Sorting Function

The sorting functionality organizes employee records using the available criteria, including employee name and net salary.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 6
```

```
----------------------------------
Enter your choice: 6

--- Sort Employees ---
1. By name (A-Z)
2. By net salary (highest first)
Choose sorting method: 1
Employees sorted successfully.
```

8. Payroll Summary Function

The payroll summary presents payroll information calculated from the employee records maintained by the program.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 7
```

```
----------------------------------
Enter your choice: 7

--- Payroll Summary Report ---
Total Employees      : 2
Full-Time Employees  : 1
Part-Time Employees  : 1
Total Gross Payroll  : PKR 74000.00
Total Deductions     : PKR 3100.00
Total Net Payroll    : PKR 70900.00
```

IMPLEMENTATION AND CODE EXPLANATION

9. Salary Statement and History Functions

The salary statement functionality generates an individual employee statement, while payroll history records payroll transactions in a text file when the relevant operations succeed.

```
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
=======================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
---------------------------------------
Enter your choice: 8
```

```
--- Process Payroll ---
Enter employee ID: 101

Employee ID : 101
Name        : Shafay
Department  : Maths
Type        : Full-Time
Gross Pay   : PKR 60000.00
Deductions  : PKR 3000.00
Net Pay     : PKR 57000.00
Basic Salary: PKR 55000.00
Bonus       : PKR 5000.00
Type YES to record this payroll: yes
Salary statement saved as salary_statement_101.txt
Payroll recorded successfully.
```

```
=======================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
=======================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
---------------------------------------
Enter your choice: 9
```

```
--- Payroll History ---
Date: 2026-10-29 14:00:30
Employee ID: 101
Name: Shafay
Department: Maths
Employment Type : Full-Time
Monthly Salary  : PKR 55000.00
Bonus           : PKR 5000.00
Gross Pay       : PKR 60000.00
Deductions      : PKR 3000.00
Net Pay         : PKR 57000.00
---------------------------------------
```

FILE HANDLING AND DATA PERSISTENCE

File handling is an important feature of the Advanced Employee Management System because it allows employee information and payroll-related records to be stored outside the program's temporary memory.

The application uses employees.txt, payroll_history.txt, and individual salary statement files.

1. **Employee Data File**

The file employees.txt stores employee records so that saved information can be retrieved during a later execution when the file is available.

The program uses a text-based record format to distinguish between full-time and part-time employees.

2. **Full-Time Employee Record Format**

A full-time employee record may resemble the following:

F|101|Sara Khan|Human Resources|50000|5000|2000

The example represents the employee category, employee ID, name, department, monthly salary, bonus, and deductions.

3. **Part-Time Employee Record Format**

A part-time employee record may resemble the following:

P|102|Ali Ahmed|Sales|500|80|3000|1000

The example represents the employee category, employee ID, name, department, hourly rate, hours worked, bonus, and deductions.

These examples illustrate the expected record structure and should be checked against the actual field order used by the program.

4. **Payroll History File**

The file payroll_history.txt stores payroll transaction information when payroll processing is recorded successfully.

This file provides a text-based record of payroll activity.

FILE HANDLING AND DATA PERSISTENCE

5. **Individual Salary Statement**

The program generates a separate salary statement for the selected employee.

For example, processing payroll for employee ID 101 may generate:

salary_statement_101.txt

The actual filename depends on the employee ID used by the application.

6. **Saving and Loading Data**

The application uses C++ file streams to read and write records.

- ifstream is used to read employee records.
- ofstream is used to write or save employee information and salary statements.
- Append mode can be used to add payroll transactions to the existing history file.

File-opening and writing checks help the application determine whether the relevant operation can proceed.

7. **Importance of File Handling**

Without file storage, employee information held only in memory may be lost when the program terminates.

Saving records to text files helps preserve employee information and payroll history when the files remain available in the working directory.

8. **Online Compiler Considerations**

Online compilers may use temporary or persistent workspaces. Successful file operations during one run do not guarantee that records will remain available indefinitely.

```
========================================
    ADVANCED EMPLOYEE MANAGEMENT SYSTEM
========================================
1.  Add Employee
2.  Display All Employees
3.  Search Employee
4.  Update Employee
5.  Delete Employee
6.  Sort Employees
7.  Payroll Summary Report
8.  Process Payroll & Generate Statement
9.  View Payroll History
10. Save Records
0.  Save and Exit
----------------------------------------
Enter your choice: 10
```

```
Enter your choice: 10
Employee records saved successfully.
```

Wrap employees.txt

```
1  F|101|Shafay|Maths|55000|5000|3000
2  P|102|Zeeshan Bashir|Mathematics|1500|6|5000|100
3
```

INPUT VALIDATION AND ERROR HANDLING

Input validation and error handling help the application respond appropriately when users enter invalid information or request operations that cannot be completed.

1. **Employee Information Validation**

Employee information is processed according to the validation rules implemented in the program. Required text and numeric fields should be checked before creating or updating an employee record.

2. **Numeric Input Validation**

The application uses input-processing functions to read numeric information such as employee IDs, salary amounts, hourly rates, hours worked, bonuses, and deductions.
Invalid numeric input should be handled according to the implemented validation logic.

3, **Duplicate Employee IDs**

The employee ID is used to identify an employee record. Where duplicate checking is implemented, an ID already assigned to another employee is rejected.

4. **Invalid Menu Selections**

The menu-driven interface processes the user's choice and directs execution to the corresponding operation. Invalid choices should be handled without causing unintended program behavior.

5. **Employee Search and Modification**

If an employee cannot be found, the program reports the unsuccessful search or modification request according to its implemented logic.

6. **Salary Input Validation**

Salary-related values should satisfy the restrictions implemented in the program. For example, negative salary amounts, bonuses, deductions, or hourly rates should be rejected if the corresponding validation checks are present.

7. **File-Handling Errors**

The program depends on the availability and accessibility of its employee and payroll files.
File-opening and file-writing checks help identify situations in which the requested operation cannot be completed.

8. **Importance of Error Handling**

Clear error messages help users understand why an operation could not be completed. Input validation and exception handling also make the program easier to test and maintain.

```
----------------------------------------
Enter your choice: a
Invalid input. Enter a whole number.
Enter your choice: 1

--- Add Employee ---
Enter employee ID: 101
An employee with this ID already exists.
```

TESTING AND RESULTS

The Advanced Employee Management System was tested through its console interface. The purpose of testing was to examine the behavior of the implemented operations and verify the results of employee management and payroll processing.

Test case	Test case	Expected Result
TC-01	Start the application	Main Menu displayed
TC-02	Add a full-time employee	A full-time employee record is created.
TC-03	Add a part-time employee	A part-time employee record is created.
TC-04	Display Employees	Stored employee records are displayed.
TC-05	Search by employee ID	The matching employee is found when the ID exists.
TC-06	Search by employee name	The matching employee is found if the name exists.
TC-07	Update employee information	The selected employee's details are updated.

TESTING AND RESULTS

The Advanced Employee Management System was tested through its console interface. The purpose of testing was to examine the behavior of the implemented operations and verify the results of employee management and payroll processing.

Test case	Test case	Expected Result
TC-08	Delete an employee	The selected employee is removed.
TC-09	Sort by name	Employee records are ordered by name.
TC-10	Sort by net salary	Employee records are ordered according to the implemented sorting logic.
TC-11	Calculate payroll	Gross pay and net pay are calculated.
TC-12	Display payroll summary	The implemented payroll summary is displayed.
TC-13	Generate salary statement	A statement file is written when the operation succeeds.
TC-14	Record payroll history	Payroll transaction information is recorded when the operation succeeds.

TESTING AND RESULTS

The Advanced Employee Management System was tested through its console interface. The purpose of testing was to examine the behavior of the implemented operations and verify the results of employee management and payroll processing.

Test case	Test case	Expected Result
TC-15	Enter invalid input	Input is handled according to the validation rules.
TC-16	Add a duplicate employee ID	The duplicate is rejected if duplicate checking is implemented.
TC-17	Save and reload employees	Saved records are restored when the environment supports persistent files.

CHALLENGES FACED AND SOLUTIONS

Developing an employee management and payroll application involves several practical considerations.

1. **Managing Different Employee Categories**

Challenge: Full-time and part-time employees use different methods of calculating base pay.

Solution: The application uses a common Employee base class and derived classes for full-time and part-time employees. Each derived class implements the payroll behavior relevant to its category.

2. **Organizing Employee Records**

Challenge: The application needs to maintain employee records while supporting additions, updates, searches, deletions, and sorting.

Solution: The EmployeeManager class organizes these operations and maintains a vector of employee objects through the common base-class interface.

3. **Applying Inheritance and Polymorphism**

Challenge: Shared employee functionality must be organized without losing the differences between employee categories.

Solution: Inheritance and virtual functions allow the program to use a common employee interface while supporting category-specific payroll calculations.

4. **Calculating Payroll**

Challenge: Gross pay and net pay depend on the employee's salary structure and the relevant bonus and deduction values.

Solution: Separate payroll calculations are provided for full-time and part-time employees according to the implemented formulas.

CHALLENGES FACED AND SOLUTIONS

5. **Searching and Sorting Records**

Challenge: Users need to locate specific employee records and organize the collection using different criteria.

Solution: Searching and sorting functions are included to support employee lookup and record organization.

6. **Maintaining Payroll Records**

Challenge: Payroll information should be available as an individual salary statement and as a history of payroll activity.

Solution: The application generates statement files and records payroll transactions in a text file when the relevant operations succeed.

7. **File Persistence**

Challenge: Employee records must be saved and retrieved, while online compiler environments may use temporary storage.

Solution: File operations were implemented using C++ file streams, and the resulting files can be inspected during testing. Persistence claims should be limited to the behavior actually verified.

8. **Input Validation**

Challenge: Invalid numeric input, duplicate identifiers, and unsuccessful search or modification requests can affect program behavior.

Solution: The program applies the validation and error-handling logic implemented in the relevant input and employee management functions.

Working through these considerations improved my understanding of object-oriented design, payroll calculations, record management, file handling, and systematic testing.

LEARNING OUTCOMES

This project provided practical experience in applying advanced C++ concepts to an employee management and payroll application.

1. **Understanding Object-Oriented Programming**

I gained practical experience in organizing employee information and payroll operations using classes and objects.

2. **Applying Encapsulation**

The project helped me understand how employee data and related operations can be organized within classes.

3. **Implementing Inheritance**

Using full-time and part-time employee classes demonstrated how derived classes can reuse common functionality from a base class.

4. **Understanding Polymorphism**

Virtual functions provided practical experience in accessing different employee types through a common interface.

5. **Working with Constructors**

The project helped reinforce the role of constructors in initializing employee objects with the required information.

6. **Managing Records with STL Vectors**

I gained experience in maintaining a dynamically sized collection of employee objects using vectors and smart pointers.

7. **Implementing Searching and Sorting**

The project provided practice in locating employee records and organizing them using different criteria.

8. Calculating Payroll

I gained experience in implementing salary calculations using monthly salaries, hourly rates, hours worked, bonuses, and deductions.

LEARNING OUTCOMES

9. **Implementing File Handling**

The project provided practical experience in saving employee records, loading saved information, generating salary statements, and recording payroll history through file streams.

10. **Understanding Input Validation and Exception Handling**

The application highlighted the importance of validating user input and responding appropriately to invalid entries and unsuccessful operations.

11. **Testing and Debugging**

Testing individual menu operations helped me examine program behavior, check payroll calculations, and verify the results of file operations.

12. **Technical Documentation**

Preparing the source code, README, screenshots, and project report provided experience in documenting a software project for review.

Overall, this task helped connect theoretical C++ concepts with practical software development and strengthened my understanding of object-oriented programming and payroll management.

CONCLUSION

The Advanced Employee Management System with Payroll was developed as part of Devixo Solutions Internship Task 04 using C++.

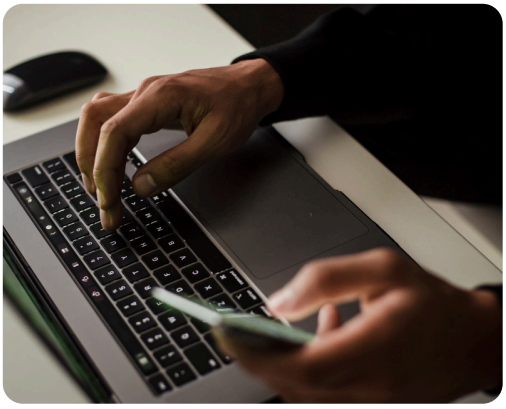
The application provides a menu-driven approach to managing employee records and includes functionality for adding, displaying, searching, updating, and deleting employees. It also supports full-time and part-time employee categories, payroll calculations, sorting, payroll summaries, salary statement generation, and payroll history.

The project demonstrates the practical application of classes, objects, encapsulation, inheritance, polymorphism, constructors, STL vectors, functions, searching, sorting, input validation, exception handling, and file handling.

Testing the application provided an opportunity to examine its main operations, review payroll calculations, and check the behavior of employee and payroll files in the development environment.

The project strengthened my understanding of object-oriented design, employee record management, payroll processing, data structures, file operations, testing, and technical documentation.

This task represents a valuable step in my learning journey as a computer science student and provides a foundation for developing more advanced business management applications in the future.



Project Summary

Items	Details
Organization	Devixo Solutions
Internship Domain	C++ Programming
Task Number	Task 04
Project Title	Advanced Employee Management System with Payroll
Programming Language	C++
Development Environment	CodeUtility online C++ compiler
Main Source File	main.cpp
Employee Data File	employees.txt
Payroll History File	payroll_history.txt
Documentation	Project report in PDF format

GITHUB REPOSITORY AND PROJECT DETAILS

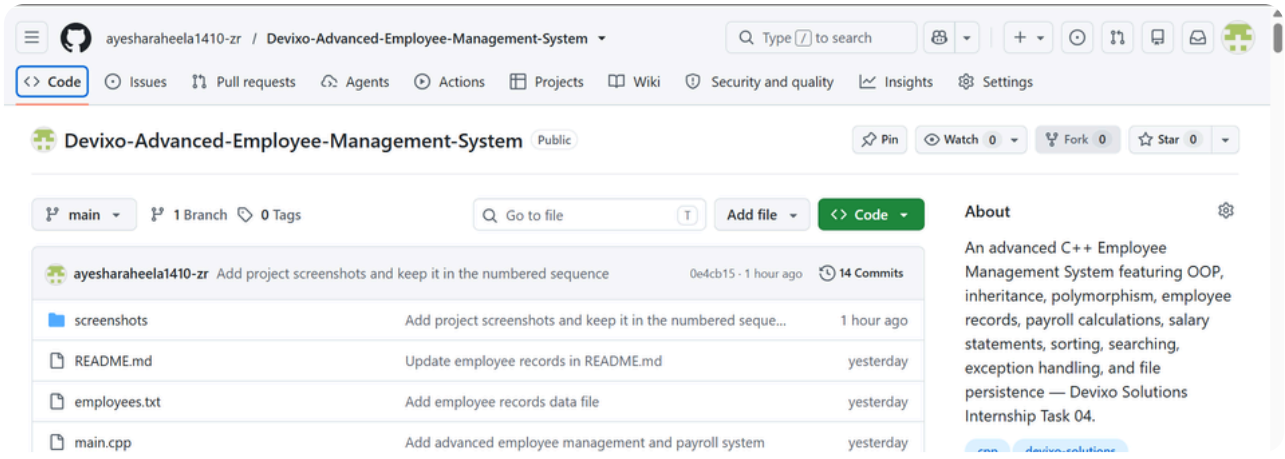
The project source code and documentation are maintained in a GitHub repository.

Repository Name: Devixo-Advanced-Employee-Management-System
Repository Link:

https://github.com/ayesharaheela1410-zr/Devixo-Advanced-Employee-Management-System

The repository contains the project source code and supporting documentation. Depending on the files uploaded, it includes:

- main.cpp — C++ source code.
- employees.txt — employee records data file.
- payroll_history.txt — payroll transaction history.
- statement.txt or salary_statement_101.txt — an individual salary statement, depending on the actual filename used.
- README.md — project overview and usage instructions.
- screenshots/ — screenshots documenting the application.
- Devixo_Employee_Management_System_Report.pdf — final project report.



FINAL STATEMENT



I am grateful for the opportunity to apply my programming knowledge to a practical project as part of the Devixo Solutions C++ Internship.
This task helped me strengthen my understanding of object-oriented programming, record management, file handling, testing, and software documentation.
The experience has encouraged me to continue developing my technical skills and work toward building more structured and reliable software applications.

Submitted by: Ayesha Raheela

End of Report

Thank You!